Hesam Shafiei Nia

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RESEARCH INTEREST

- Robust, Uncertainty-Aware Machine Learning from Sparse and Biased Observational Data
- Deep Learning and Generative Modeling for Spatio-temporal Trajectory Data
- Control Theory and Optimization for Decision-Making in Transportation Systems
- Data-Driven Modeling and Control of Large-Scale Transportation and Aviation Networks
- Sustainable Transportation and Aviation: Emissions Estimation and Generative Trajectory Modeling for Environmental Impact

EDUCATION

M.Sc. in Transportation Engineering *Sep. 2024 - Present*

University of Washington, Department of Civil & Environmental Engineering, Seattle, WA

(Thesis Defense Expected 2026)

Thesis Advisors: Dr. Don MacKenzie & Dr. Cynthia Chen

Thesis (in progress): Generated Trajectory-based Emission Modeling for Operational Decision Support GPA: 3.63 / 4.0

M.Sc. in Transportation Engineering *Sep. 2017 – Sep. 2020*

Islamic Azad University Science and Research Branch (SRBIAU), Department of Civil and Environmental Engineering, Tehran, Iran

Thesis: Traffic and Environmental Simulation of the Effects of Increasing Pedestrian Infrastructure and Public Transportation on the Central Area of the City Using AIMSUN Software (Case Study: Sarcheshmeh Square to Shahid Modares – Imam Khomeini Intersection in Ardabil)

Advisors: Dr. Seyed Mohammad Seyed Hosseini & Dr. Hassan Javanshir

GPA: 16.60 / 20

B.Sc. in Civil Engineering *Sep. 2013 – Jul. 2017*

Ilam University, Department of Civil and Environmental Engineering, Ilam, Iran

PUBLICATIONS

*(Published as **Hesam Shafienya**)*

Peer Reviewed Journal Articles

- **Shafienya**, **Hesam**, Amelia C. Regan. “4D Flight Trajectory Prediction Using a Hybrid Deep Learning Prediction Method Based on ADS-B Technology: A Case Study of Hartsfield–Jackson Atlanta International Airport (ATL).” Transportation Research Part C: Emerging Technologies 144 (2022): 103878.
- **Shafiei Nia**, **Hesam**, Don MacKenzie. "3-D Emissions Mapping and Social Cost Estimation for US Domestic Aviation at West Coast Hubs." Under review; preprint: arXiv:2609.31686 (2026). https://arxiv.org/abs/2609.31686

Conference Papers & Proceedings

- **Shafienya**, **Hesam**, Amelia C. Regan. “4D Flight Trajectory Prediction Based on ADS-B Data: A Comparison of CNN-GRU Models.” 2022 IEEE Aerospace Conference (AERO). IEEE, 2022.
- **Shafiei Nia**, **Hesam**, Don MacKenzie “Trajectory-based Emission Modeling for Operational Decision Support.” (INFORMS 2026 Annual Meeting, accepted.)
- **Shafienya**, **Hesam**, Kamran Rahimov. “Safety Analysis Effects of Push-Bottom with Countdown Timer on None-Motorize Traffic in Intersections.” First National Conference on Sustainable Development in Civil Engineering, Architecture and Urbanism, Tehran, Iran, 2017. (In Persian)

Presentations

- **Shafienya**, **Hesam**, Amelia C. Regan. 4-D Flight Trajectory Prediction: Comparison of CNN-GRU Model Structure Based on ADS-B Data. IEEE AeroConf, Big Sky, MT, 2022
- **Shafienya**, **Hesam**, Hassan Javanshir. Imperialist Competitive Algorithm (ICA): An Algorithm for Optimization Based on Imperialistic Competition. Allameh Tabataba'i University, Tehran, Iran, 2018.

PROJECTS

Generating Flight Trajectories Using Variational Autoencoders for Emissions Mapping and Social Costs Estimation *Dec. 2025 - Present*

Supervisors: Dr. Don MacKenzie, Dr. Cynthia Chen, University of Washington, Seattle, WA

Designing autoencoder-based generative models to synthesize realistic flight trajectories, coupled with downstream emissions mapping to support social costs analysis for air transportation.

ADS-B Trajectory Data Imputation and 3D Spatial Emissions Mapping

Supervisors: Dr. Don MacKenzie, University of Washington, Seattle

Reconstructed ADS-B trajectory data using an autoencoder-based imputation model, then developed a 3D spatial emissions mapping framework using BADA (Base of Aircraft Data) aircraft performance models to estimate fuel burn and emissions along flight trajectories.

4-D Flight Trajectory Prediction – Potential Operational and Environmental Impacts of Improved Machine Learning Models *May 2021 – Jul. 2022*

Supervisor: Dr. Amelia Regan, University of California, Irvine

Traffic and Environmental Simulation of the Effects of Increasing Pedestrian Infrastructure *Sep. 2018 – Sep. 2020*

Supervisor: Dr. S. M. Seyyed Hosseini, Islamic Azad University Science and Research Branch (SRBIAU), Tehran, Iran

EXPERIENCE

Graduate Research Assistant — Heavy- and Medium-Duty Vehicle (HDV/MDV) Electrification & Emissions Research (The State of Zero-Emissions Delivery in the U.S) *Fall 2024*

Urban Freight Lab (UFL), University of Washington, Seattle

Supervisor: Kelly Rula

Researching electrification, charging infrastructure, and emissions implications for heavy- and medium-duty vehicle fleets.

Research Assistantship— Supported research in transportation systems modeling and optimization *Nov. 2020 – Oct. 2021; Apr. 2023 – May 2023*

Islamic Azad University Science and Research Branch (SRBIAU)

Supervisor: Dr. S. M. Seyyed Hosseini, Dr. Hassan Javanshir

Microscopic traffic simulation (AIMSUN) and the analysis of traffic and environmental impacts of pedestrian and public transportation infrastructure.

TEACHING EXPERIENCE

Teaching Assistant, *Sep. 2024 – June 2026*

University of Washington

- SCTL 508: Collaboration Across the Supply Chain *Fall 2024*
  Instructor: William M. Keough

- SCTL 510/ CET 591: Freight Transport *Winter 2025*
  Instructor: Dr. Tom Maxner

- SCTL 503: Inventory Management *Winter 2025*
  Instructor: Apurva Jain

- SCTL 502: IT Systems and Supply Chain Integration *Spring 2025*
  Instructor: William M. Keough

- SCTL 506: Finance and Performance Management *Spring 2025*
  Instructor: Dan Stull
- CEE 416: Planning for People and Freight. *Fall 2025*
  Instructor: Dr. Travis Fried

- CET 513: Transportation Networks and Optimization *Fall 2025*
  Instructor: Dr. Jeff Ban

- CET 581: Travel Demand Forecasting (led Python instruction) *Winter 2026*
  Instructor: Dr. Cynthia Chen

- CEE 410: Traffic Engineering Fundamentals (led simulation lab instruction) *Winter 2026*
  Instructor: Dr. Jeff Ban

- *CEE 220 Introduction to Mechanics of Materials (led labs instruction)* *Spring 2026*
  Instructor: Dr. Peter Mackenzie-Helnwein

Teaching Assistant, *Sep. 2018 – Jan. 2019*
Islamic Azad University Science and Research Branch (SRBIAU), Tehran, Iran

- Soft Computing *Fall 2018*
  Instructor: Dr. Hassan Javanshir

OTHER TEACHING INTERESTS

- Supply Chain Transportation and Logistics
- Programming in Python
- SQL
- Introduction to AI & ML
- Deep Learning

LANGUAGES

- English (Professional – Writing and Spoken)
- Persian (Native)

SOFTWARE

- Programming Tools: Python, SQL, R, MATLAB, Git/ GitHub, Linux, Latex, Docker, Bash
- Transportation Simulators: VISSIM, Synchro, HCS (Highway Capacity Software), AIMSUN
- Python Libraries: PyTorch, TensorFlow, Keras, scikit-learn, NumPy, Pandas, SciPy, Matplotlib
- Graphics: Visio, NCSS